DOC2PPT: Automatic Presentation Slides Generation from Scientific Documents

A generated deck - draft, not final

Abstract

- We propose a hierarchical sequence-to-sequence approach to tackle our task in an end-to-end manner
- Can machines learn to emulate this laborious process?
- Our approach exploits the inherent structures within documents and slides and incorporates paraphrasing and ...
- We show that our approach outperforms strong baselines and produces slides with rich content ...
- We present a novel task and approach for document-to-slide generation
- Creating presentation materials requires complex multimodal reasoning skills to summarize key concepts and arrange ...

Introduction

- We hope that our DOC2PPT will advance the state-of-the-art in the vision-and-language domain
- In this paper, we introduce DOC2PPT, a novel task of creating presentation slides from ...
- This requires solving several challenges in the vision-and-language domain, e.g., visual-semantic embedding and multimodal ...
- Lastly, we explore both template-based and learning-based layout design and compare them both quantitatively ...
- It requires skills to abstract complex concepts and conveys them in a concise and ...
- Furthermore, none of the above addresses the challenge of find-ing an optimal visual layout ...

Related Work

- We propose a large dataset of 5,873 pairs of high-quality scientific documents and slide ...
- Learning text-image similarity has been studied in the visual-semantic embedding (VSE) literature (Karpathy and ...
- As argued above, our task brings a new suite of challenges to vision-and-language understanding
- Furthermore, existing datasets in this domain (such as TalkSumm (Lev et al
- Joint modeling of vision-andlanguage has been studied from different angles
- Our task involves generating slides with relevant text and figures

Figure 1: We introduce DOC2PPT, a novel task of gener-

Approach

- Table 1 reports the descriptive statistics of our dataset
- We tune the two hyper-parameters θR and θA via cross-validation (we set θR = ...
- For simplicity, we discard F out if its highest visual embedding similarity is lower ...
- The goal of DOC2PPT is to generate a slide deck from a multimodal document ...
- Figure 2: An overview of our architecture
- This is a natural choice to encode our prior knowledge about the hierarchical structure

Experiments

- DOC2PPT is a new task with no established evaluation metrics and baselines
- We carefully ablate various components of our approach and evaluate them on our proposed ...
- We propose automatic metrics specifi- cally designed for evaluating slide generation methods
- We also perform hu-man evaluation to assess the generation quality

Evaluation Metrics

- Each layer of the PT is based on a 256-unit GRU
- Implementation Detail For the DR, we use a Bi-GRU with 1,024 hidden units and ...
- The PAR is designed as Seq2Seq with 512-unit GRU
- Therefore, we propose the slide-level ROUGE: ROUGE-SL = ROUGE-L × e |Q–˜ Q| Q ...
- Intuitively, the number of slides in a deck is a good proxy for the ...
- Ablation Settings ROUGE-SL LC-FS TFR mIoU Hrch-PT PAR TIM Post Proc

Results and Discussions

- Table 2 (a) and (b) compare the two models on the four metrics
- Table 2 (f) shows this post-processing further improves the two image-related metrics, LC-FS and ...
- Significance tests: two-sample t-test ($p < 0.05$.) maining space under the main content
- How well do the figure(s)/tables(s) in DECK A match the text or figures/tables in ...
- The responses were all on a scale of 1 (not similar at all) to ...
- The workers were then asked to answer three questions: Q1

Conclusion

- Our slide stemming step is 93% accurate based on the human annotated validation set
- Sentence Matching Table 4 shows examples of matching sentences between the paper and the ...
- Precision, recall, and F1 are evaluated using the human-labeled testing set
- Slides are matched with the section with the highest cosine similarity and our slide-section ...
- Also, using hobj benefits more (w/ 32.27 vs w/o 31.95)
- Details of the Data Processing Steps Section 4 in our main paper explains how ...